

Session 8



Trainee Program: Develop Greatness

Software Dev Final Project

Saturday, April 25th.



Agenda

01 Timeline

02 Project Requirements

03 Evaluation Criteria

04 Q&A

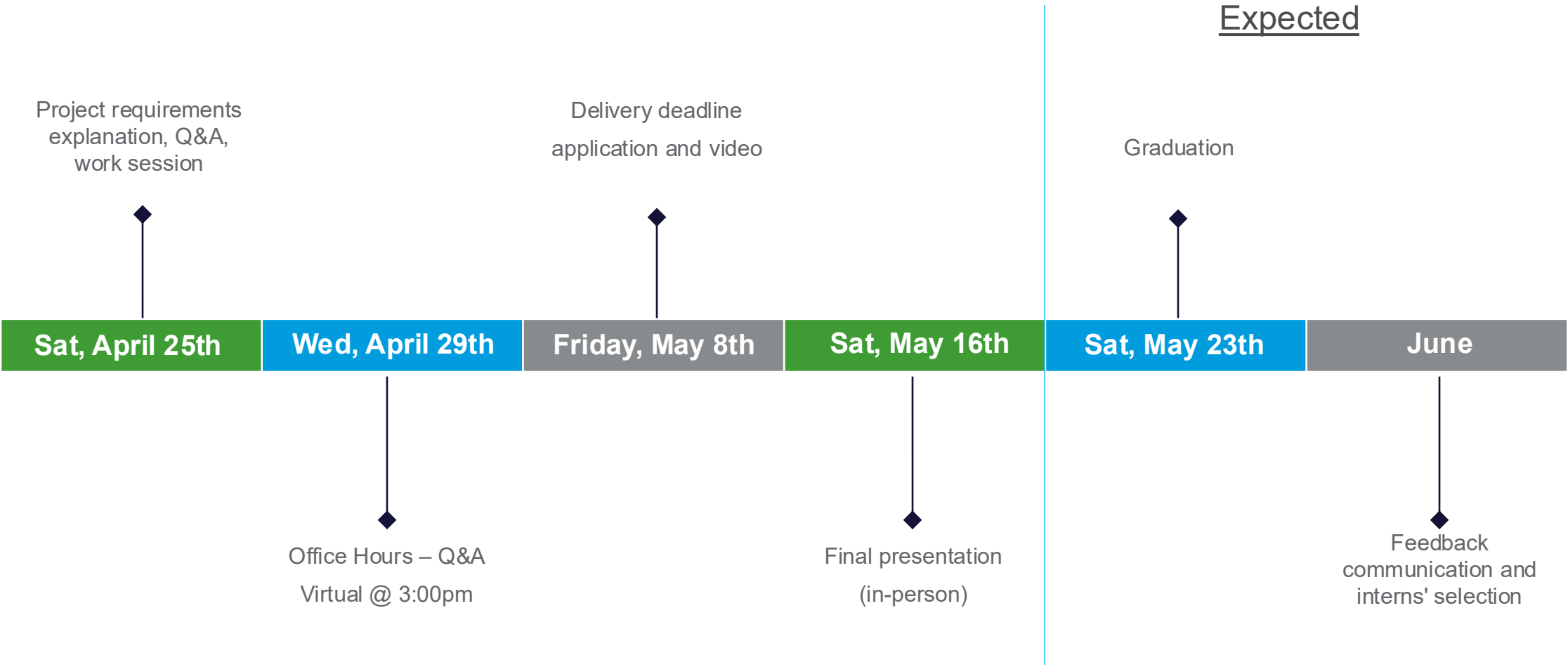
05 Work Session



RSM

Timeline

Final Project Timeline



Requirements

Requirements overview

Northwind Traders, a global shipping company, requires a modern Order Management System integrating geo-location services to enhance operational efficiency. The system should enable employees to process orders with real-time address validation, visualize shipment locations, and generate PDF reports for order tracking.

Requirements

1. Order Management (CRUD Operations)

Form Fields: Customer selection, order date, shipping details (validated address), product/quantity selection, employee assignment.

Address Validation:

- Integrate Google Maps Address Validation API to standardize addresses.
- Display validated addresses with geocoded coordinates (latitude/longitude).

2. Google Maps Integration

Embedded Map: Show validated addresses on an interactive map during order creation/update.

Geocoding: Convert addresses to coordinates for shipment tracking.

3. Reporting

- **Graphics:** One or two graphics, such as **order per time** and **shipments by region**.
- **Order Details Table**, displaying key order and shipment fields (customer, order date, products, region, and totals). Users must be able to filter order by year/month/week by region, the table header

should include toolbar actions to export data to Excel **and** PDF.

- **Order Details Report:** Export order metadata (dates, freight charges) and line items to a styled PDF with company branding.

Technical Specifications

Database Layer

Use Entity Framework Core to map Orders and Order Details tables from the Northwind database.

Include relationships to Customers, Employees, and Shippers.

Backend

C#: Implement using ASP.NET Core (REST API).
Use CLEAN Architecture.

Frontend

Vue3 + Quasar Framework, (no TS required)

Requirements

Testing

Backend Unit Tests: Use AI tools (GitHub Copilot) to generate tests for API endpoints and business logic.

Bonus: Frontend tests for form validation and map interactions.

Deliverables

1. **Order Form:** CRUD interface with embedded Google Maps.
2. **Report Generation:** Dashboard showing basic shipment and order insights using 1-2 visuals and a report table.
3. **Test Suite:**
 - Backend: 80% coverage for services/controllers.
 - Bonus: Frontend tests for address validation workflow.
4. **Documentation:** Setup guide for API keys and database configuration.

<https://github.com/microsoft/sql-server-samples/blob/master/samples/databases/northwind-pubs/instnwnd.sql>

<https://www.youtube.com/watch?v=YHm4gWJyoyA>

Bonus Points

Implementing frontend-to-backend error handling.

App in Docker Container

UI Sketch – Expected UI Sample

Order #10248

New

Save

Delete

GENERATE INVOICE

ORDER INFORMATION

Customer

Alfreds Futterkiste

Shipping Address

Obere Str. 57, 12209 Berlin, Germany

Order Date

10/27/2023

Account Manager

Andrew Fuller

Line Items

3 ITEMS

ADD LINE

DELETE SELECTED

PRODUCT DESCRIPTION	QUANTITY	UNIT PRICE	EXT. TOTAL
Chai Beverages 10 boxes x 20 bags	12	\$18.00	\$216.00
Chang Beverages 24 - 12 oz bottles	10	\$19.00	\$190.00
Aniseed Syrup Condiments 12 - 550 ml bottles	5	\$10.00	\$50.00
ORDER TOTAL			\$456.00

VALIDATED LOGISTICS ADDRESS

VERIFIED

STREET

Obere Str. 57

CITY

Berlin

STATE / PROV.

Berlin (BE)

POSTAL CODE

12209

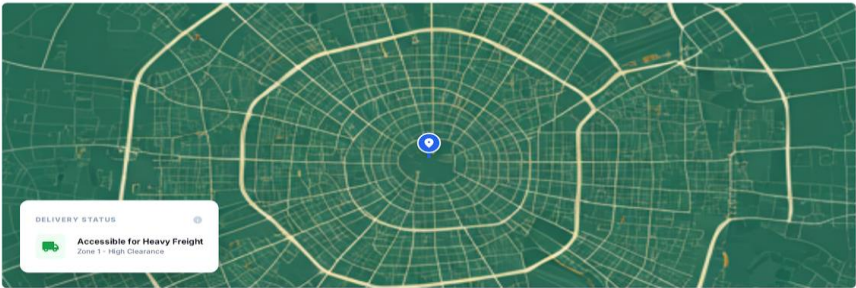
COUNTRY

Germany

GPS COORDINATES

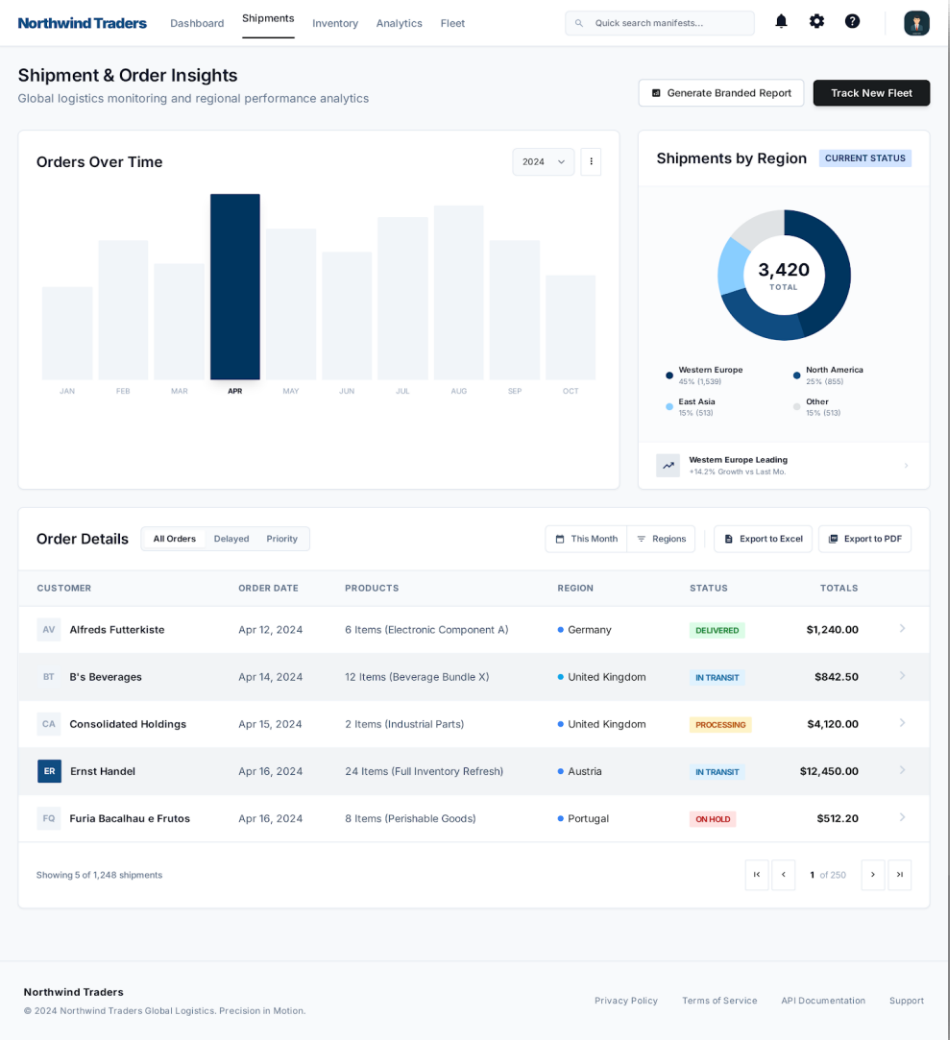
52.5290° N, 13.4050° E

Help Desk



DELIVERY STATUS

Accessible for Heavy Freight
Zone 1 - High Clearance



Order Details

All Orders

Delayed

Priority

This Month

Regions

Export to Excel

Export to PDF

CUSTOMER	ORDER DATE	PRODUCTS	REGION	STATUS	TOTALS
AV Alfreds Futterkiste	Apr 12, 2024	6 Items (Electronic Component A)	Germany	DELIVERED	\$1,240.00
BT B's Beverages	Apr 14, 2024	12 Items (Beverage Bundle X)	United Kingdom	IN TRANSIT	\$842.50
CA Consolidated Holdings	Apr 15, 2024	2 Items (Industrial Parts)	United Kingdom	PROCESSING	\$4,120.00
ER Ernst Handel	Apr 16, 2024	24 Items (Full Inventory Refresh)	Austria	IN TRANSIT	\$12,450.00
FQ Furia Bacalhau e Frutos	Apr 16, 2024	8 Items (Perishable Goods)	Portugal	ON HOLD	\$512.20

Showing 5 of 1,248 shipments

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<

1 of 250

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Northwind Traders

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Privacy Policy

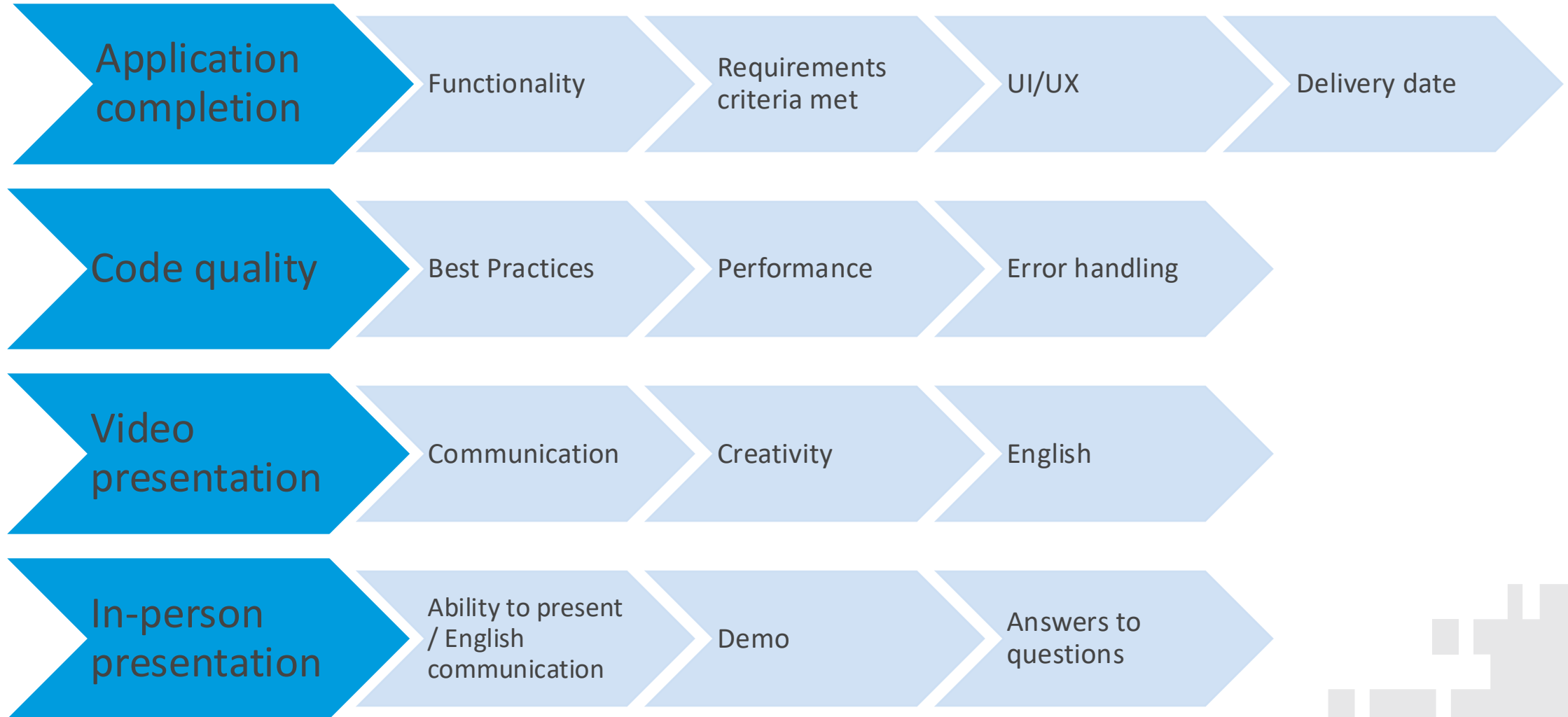
Terms of Service

API Documentation

Support

Evaluation Criteria

Evaluation Criteria



Considerations for Video

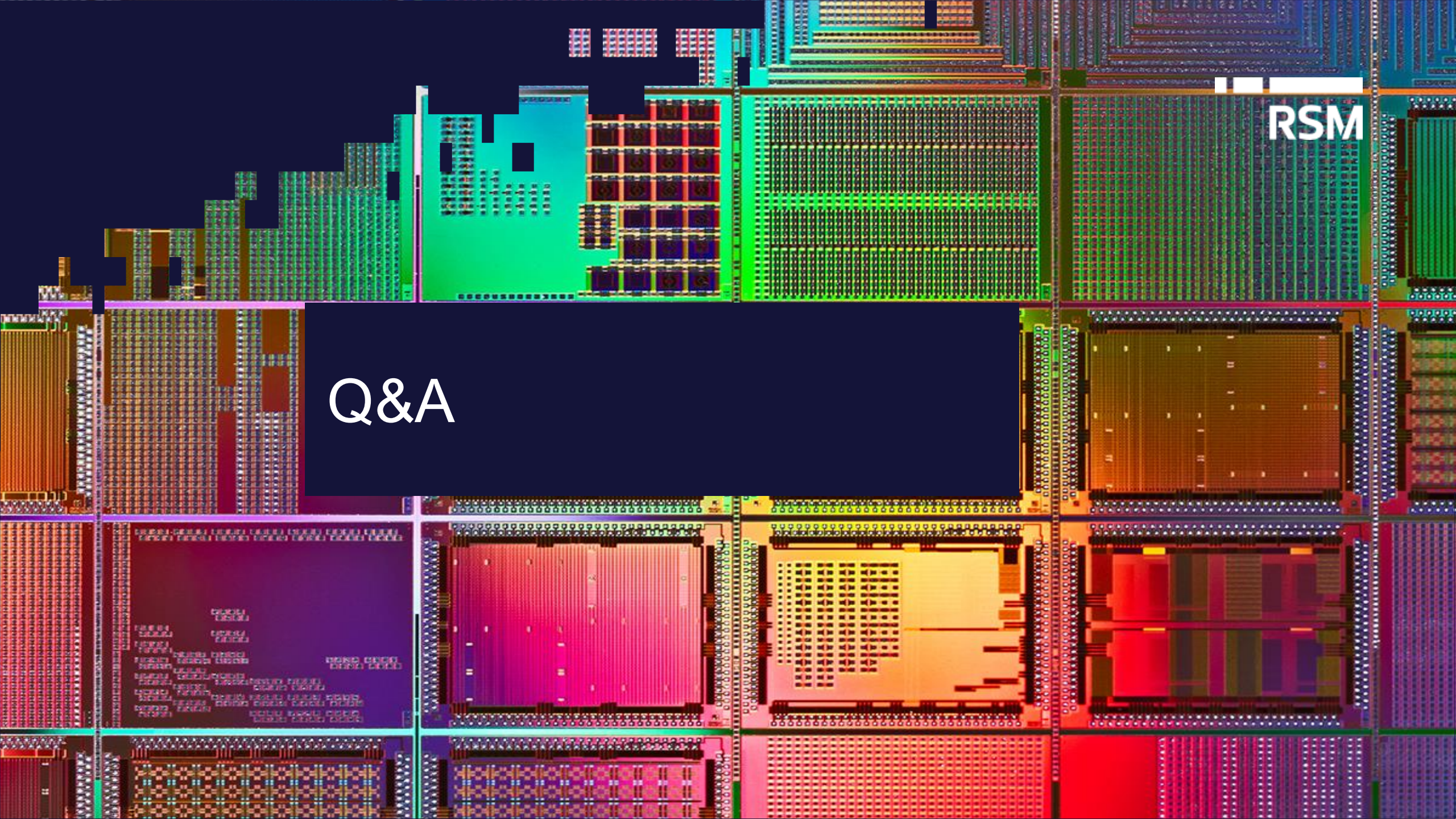
- Time: max. 4 minutes

Deliverables

- Video: upload it to the Moodle assignment
- Project: share the Git repository, add the link to your repository in the Moodle deliverable

Considerations in-person presentation

- Demo in English: explain the solution, how it works/usage
- Code review: explain the most relevant components and logic of your project
- Time: 10 minutes + 5 minutes for Q&A (You should be able to answer questions about your project)



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Q&A



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Work session
Let's get to it!

We value your feedback! Please submit your feedback it help us to improve your experience

